CONSTRUCTION SPECIFICATIONS

GENERAL: ALL MATERIALS SHALL BE INSTALLED IN A WORKMANLIKE MANNER AND THE METHOD BE WITHIN THE HIGHEST STANDARDS OF THE APPLICABLE CONSTRUCTION TRADE. ALL WORK SHALL MEET OR EXCEED PROVISIONS OF ALL GOVERNING CODES AND ORDINANCES. THE DRAWINGS ARE GENERAL IN NATURE. IT IS INTENDED THAT THE CONTRACTOR ADJUST DIMENSIONS, MATERIALS AND METHODS OF CONSTRUCTION TO PROPERLY ADAPT THE WORK TO THE SITE CONDITIONS.

EXCAVATOR: EXCAVATE FOR ALL FOUNDATION WORK TO DEPTH BELOW GRADE REQUIRED BY LOCAL CODE. FOOTINGS SHALL REST ON UNDISTURBED EARTH.

CONCRETE: SOLID CONCRETE MASONRY UNITS OR 2500# STRENGTH CONCRETE OF A SIZE SUFFICIENT TO SPREAD THE IMPOSED FOUNDATION LOAD ON THE BEARING SOIL IN ACCORDANCE WITH CODE REQUIREMENTS.

CARPENTRY: ALL WOOD SHALL BE #2 OR BETTER SOUTHERN YELLOW PINE, PRESSURE–TREATED WITH NON–ARSENIC BASED PRESERVATIVE, OF NOMINAL SIZES DETAILING INDICATED ON THE DRAWINGS OR BY CODE.

METALS: ALL HARDWARE SHALL BE HOT–DIPPED GALVANIZED OR OTHER NON–CORROSIVE FINISH AND SHALL BE OF A SIZE AND SPACING SUFFICIENT TO CARRY OR TRANSMIT THE INTENDED STRUCTURAL LOAD IN ACCORDANCE WITH GOVERNING CODES AND APPLICABLE STRUCTURAL STANDARDS. FINISH HARDWARE SHALL BE NON–CORROSIVE AND SUITABLE FOR EXTERIOR APPLICATIONS.

FLASHING: FLASHING SHALL BE NON–ALUMINUM AND APPROPRIATE TO THE TYPE OF PRESSURE TREATED WOOD USED. ADHESIVE BACKED FLASHING SHALL MEET AAMA711.

ANALYSIS OF 2 X 8 CANTILEVERED DECK JOISTS

CALCULATIONS ARE BASED ON THE FOLLOWING DATA:
- 60 PSF LIVE LOAD
- 10 PSF DEAD LOAD
- #2 SYP, Fb (ALLOWABLE BENDING STRESS) 907 PSI
- E = 1.1 E6 PSI
- CANTILEVER LENGTH 3'–0" MAX.
- BASE SPAN 8'–0" TO 9'–0"
- JOIST SPACING 16" OC
- DEFLECTION LIMITED TO L/240

CALCULATED BENDING STRESS = 654 PSI < 907 PSI OK
DEFLECTION = 0.07" = L/1402 FOR 8'–0" BASE SPAN
DEFLECTION = 0.26" = L/503 FOR 11'–0" BASE SPAN

THE CALCULATIONS INDICATE SUCH AN APPLICATION IS ACCEPTABLE AND SATISFIES THE DESIGN SPECIFICATIONS OF THE WESTERN WOOD PRODUCTS ASSOCIATION.

UPLIFT AND UPLIFT RESISTANCE CALCULATIONS

FOR THESE CALCULATIONS, THE "WORST CASE" COLUMN (LONGEST SPAN AND LARGEST CONTRIBUTING AREA) WAS USED.THE DECK WAS CONSIDERED A FLAT ROOF. TABLE 1609.6.2.1(1) OF THE IBC IS FOR A ROOF WITH MEAN HEIGHT OF 30 FEET LOCATED IN EXPOSURE B. TABLE R301.2(3) OF THE IRC LISTS THE LOWEST MEAN ROOF HEIGHT AT 15' WITH AN ADJUSTMENT COEFFICIENT OF 1.00. THIS DECK IS ±5'–4"ABOVE GRADE. DESIGN WIND SPEED = 115 MPH.

UPLIFT
DECK CONTRIBUTING AREA= 6'–0" X 8'–0" = 48SF

UPLIFT WAS INTERPOLATED FROM TABLE 1609.6.2.1(1), VERTICAL LOAD, INTERIOR ZONE, WINDWARD ROOF, LOAD DIRECTION – LONGITUDINAL.
110MPH = –16.0 120MPH = –19.1 115MPH = –17.55
48SF X 17.55PSF =842.4# UPLIFT

RESISTANCE TO UPLIFT

THE GOLIATH METAL POST HAS AN ALLOWABLE WORKING LOAD FOR UPLIFT RESISTANCE OF +/– 4800#.

DEAD LOAD 48.0SF X 10#/SF =480.0#

GOLIATH TECH METAL POST	4800	
DEAD LOAD	480	
HEADLOK	640	TOTAL = 5920 OK



3 5/8" LEDGERLOK BOLTS SPACED – STAGGERED– AS PER MFG. TABLE BELOW.

LEDGER BOLT PATTERN DETAIL NTS

BASED ON 60#LL, S. PINE LEDGER AND 2x LUMBER HOUSE RIM JOIST

SPAN FROM LEDGER TO JOIST	ON CENTER SPACING
6'–0" OR LESS	17
+6'–0" TO 8'–0"	13
+8'–0" TO 10'–0"	10
+10'–0" TO 12'–0"	8
+12'–0" TO 14'–0"	7
+14'–0" TO 16'–0"	6

THE GOLIATH TECH METAL POST INSTALLER WILL PROVIDE A FIELD REPORT DETAILING AND ATTESTING TO THE LOCATION OF EACH HELICAL PILE, IT'S DEPTH, THE ACHIEVED TORQUE AND THE BEARING CAPACITY IN RELATION TO THE TORQUE.

A COPY OF THIS REPORT WILL BE PROVIDED TO THE CONSTRUCTION OFFICIAL.

GENERAL INFORMATION

USE GROUP R–5
CONSTRUCTION TYPE 5B

DESIGN LOADS

LIVE LOAD	50#/sf
DEAD LOAD	10#/sf
TOTAL LOAD	60#/sf

DESIGN WIND SPEED = 115mph

AREA= 376SF NEW DECK

BUILDING CODES
N.J.A.C. 5:23-6
2018 International Residential Code NJ Ed.

ALTERNATE CONCRETE FOOTING SIZES (SEE NOTES BELOW)

LOAD IN LBS	DIAM
0–2137#	14"ø
2138–2799#	16"ø
2800–3533#	18"ø
3534–4362#	20"ø
4363–4809#	21"ø
4810–5279#	22"ø
5280–5770#	23"ø
5771–6283#	24"ø
6284–6817#	25"ø
6818–7374#	26"ø

NOTE:
1. VARIOUS UNMOVABLE OBSTACLES SUCH AS SEPTIC LINES, LARGE ROCKS, ETC., MAY PRECLUDE A HELICAL PIER. A CONCRETE FOOTING MAY BE SUBSTITUTED FOLLOWING THE TABLE ABOVE.
2. FOOTINGS ARE 8" DEEP @ MIN. 3'–0" BELOW GRADE. CONTRACTOR SHALL HAVE OPTION TO USE ALTERNATE CONCRETE FOOTINGS OVER HELICAL PILES

NJ License # AI 13424

Rev. No.: Date:

Jeffrey A. Fleisher, AIA Architect
56 Main Street Suite 3B
Flemington, New Jersey 08822
Tel: 908-782-5382 Fax: 908-782-5204

PROJECT TITLE
NEW DECK FOR
ALI JAMALI
BLOCK LOT
90 ORCHARD AVE
METUCHEN, NEW JERSEY

DRAWING TITLE
NOTES

Project No.	2001–71
Drawn By:	M.G. / R.E
Checked By:	J.A.F.
Date:	11 JUN. '20

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Sheet No.

FOOTING LEGEND

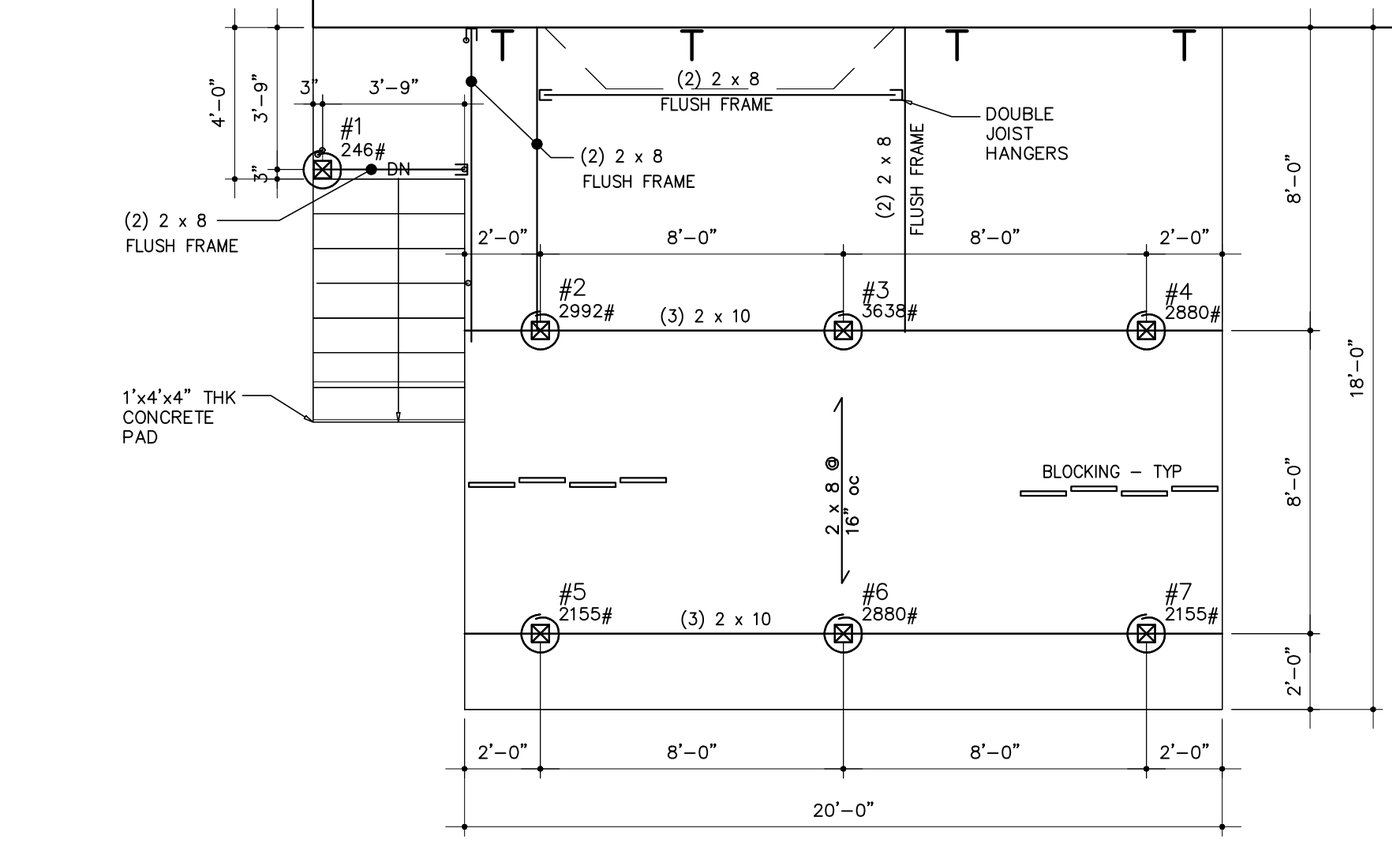
(7) POSTS – P2
(7) SUPPORT PLATE – MHC1161–3B POST CAP
HELIX SIZE – 10"

#1 FOOTING NUMBER

500# REQUIRED LOAD IN POUNDS

6 X 6 COL ON HELICAL PIER

T SIMPSON DDT1Z DECK TENSION TIES
W/SIMPSON TITEM HD SS .375
SCREWS OR EQUIVALENT



FRAMING PLAN 1/4" = 1'–0"

1. VERIFY ALL DIMENSIONS PRIOR TO CONSTRUCTION.
2. ALL NEW FOOTINGS SHALL BE AS NOTED ON PLAN.
3. PROVIDE JOIST HANGERS AT ALL FLUSH FRAME CONDITIONS.
4. BLOCK JOISTS AT MID–SPAN FOR SPANS OF 8'–0" OR GREATER.
5. 2X8 PT JOISTS @ 16"
6. 6X6 PT COLS TREATED FOR STRUCTURAL GROUND BURIAL MCA .23.

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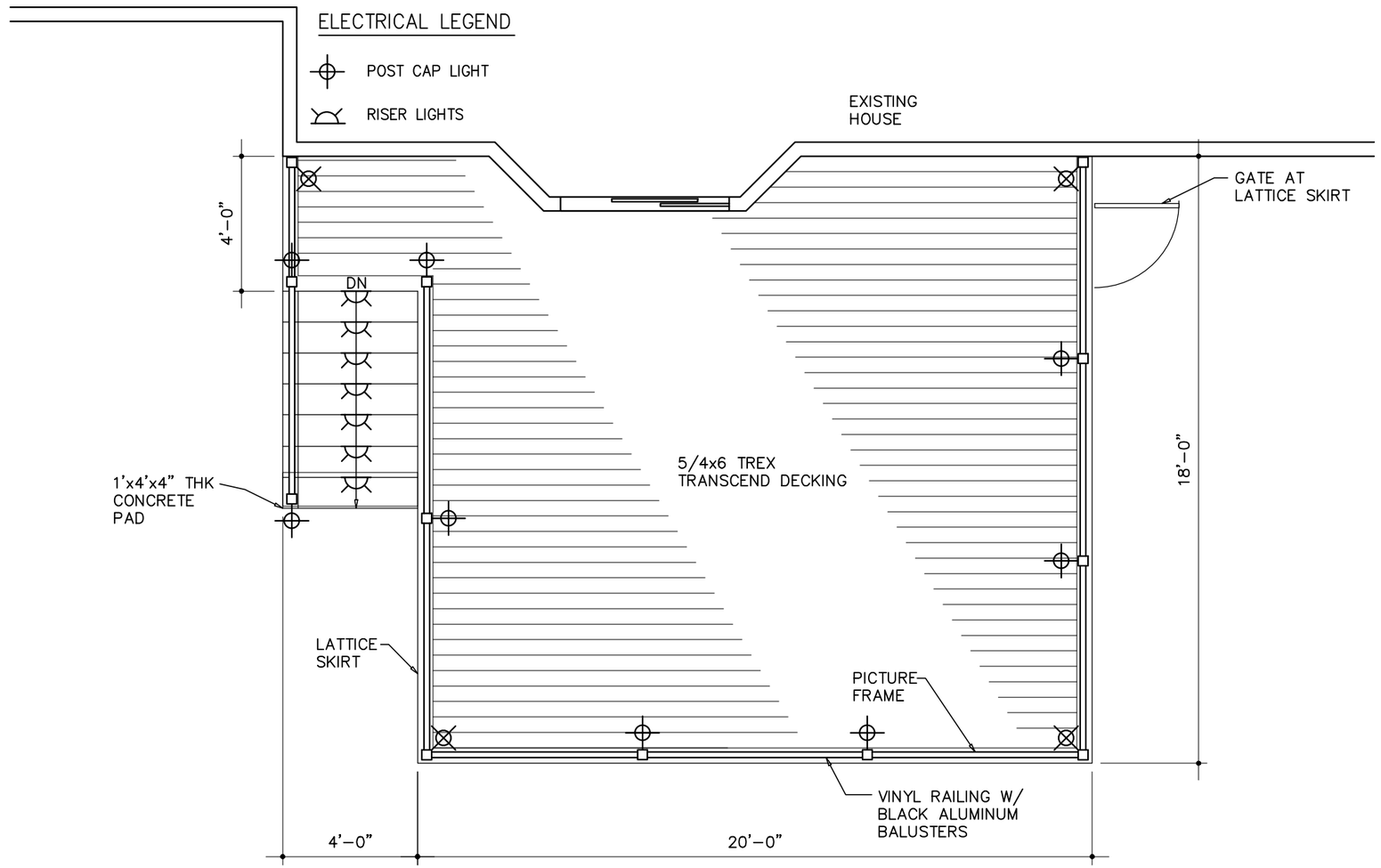
Rev. No.: Date:

PROJECT TITLE
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DRAWING TITLE
STRUCTURAL PLAN

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DECK PLAN 1/4" = 1'-0"

1. FASTEN 5/4 X 6 DECKING W/ HIDDEN FASTENERS.
2. PROVIDE VINYL RAILINGS W/ BLACK ALUMINUM BALUSTERS

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56 Main Street Suite 3B
Flemington, New Jersey 08822
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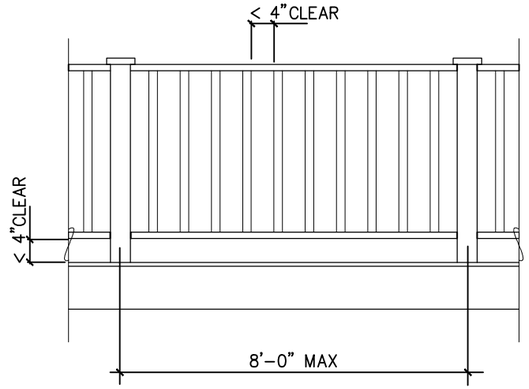
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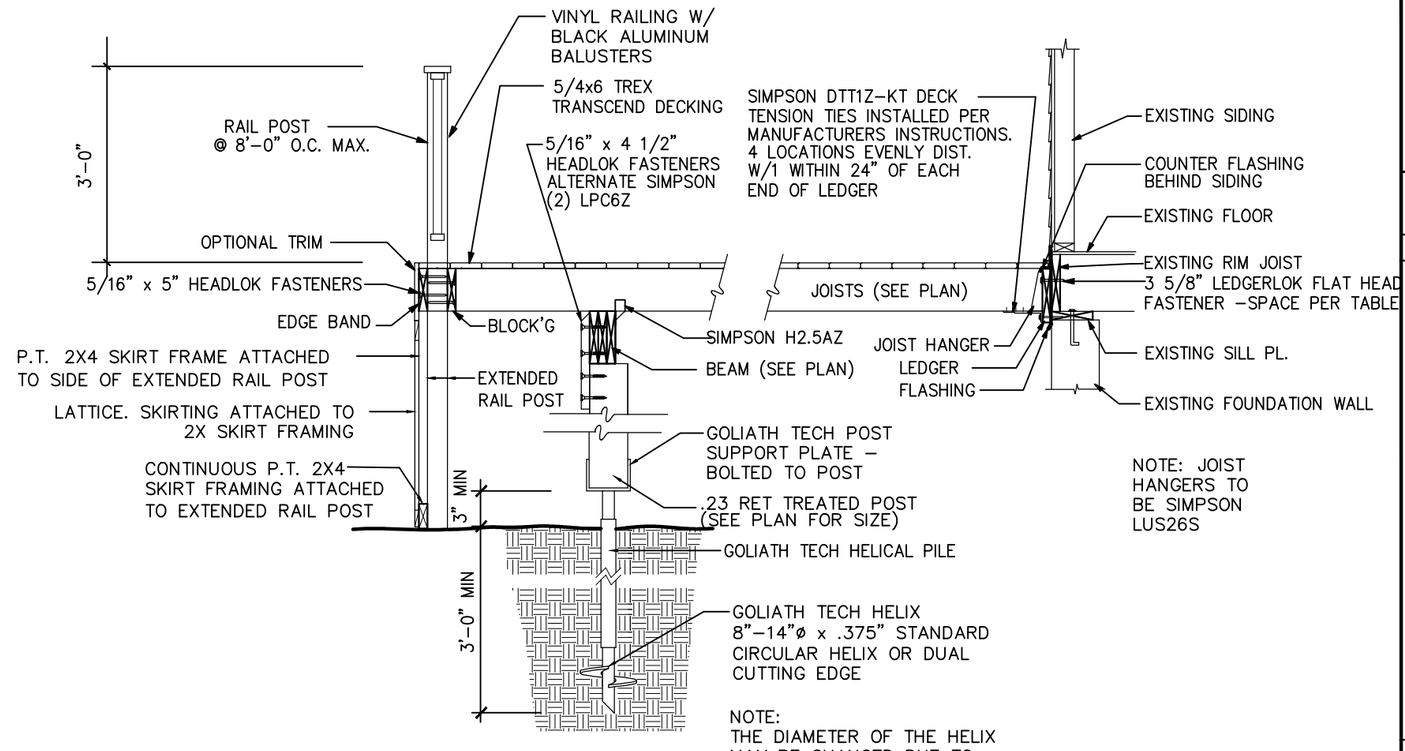
DRAWING TITLE
DECK PLAN

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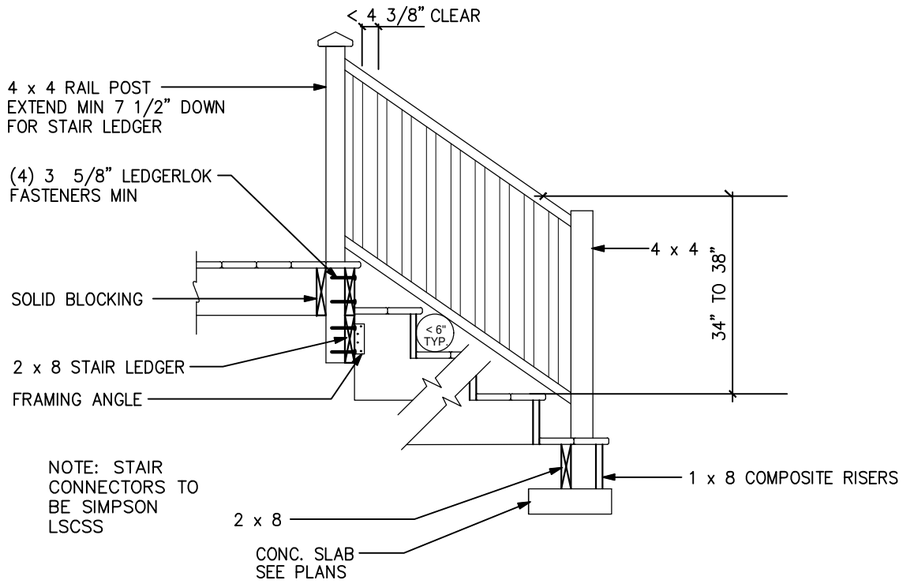
RAILING ELEVATION



NOTE: JOIST HANGERS TO BE SIMPSON LUS26S

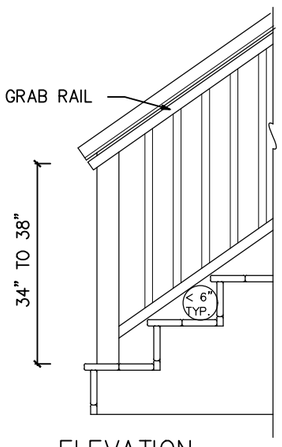
NOTE:
THE DIAMETER OF THE HELIX MAY BE CHANGED DUE TO SOIL CONDITIONS AT THE SITE. THE INSTALLER IS RESPONSIBLE FOR DETERMINING THE PROPER HELIX SIZE BASED ON SOIL CONDITIONS.

TYPICAL SECTION NTS

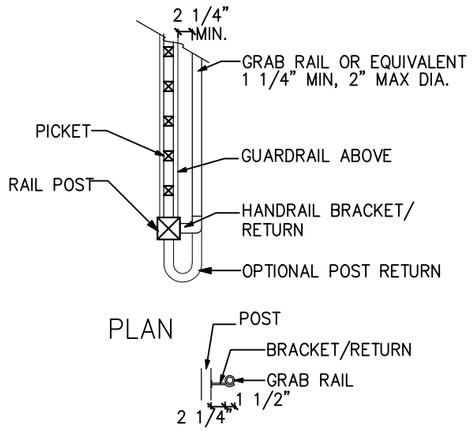


RISERS 8 1/4" MAX.
TREADS 9" MIN.

STAIR DETAIL NTS



ELEVATION



PLAN

DETAIL

GRASPABLE HANDRAIL NTS

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